THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 12: **How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A food web is: 1

(a) many interlinked food chains **(b)** a single chain
(c) a plant **(d)** a type of spider web

2. Rhizobium bacteria in root nodules of pulses help in: 1

(a) pollination **(b)** fixing nitrogen **(c)** making sugar **(d)** producing oxygen

3. Green plants are producers because they: 1

(a) make food by photosynthesis **(b)** do not need sunlight
(c) break down dead matter **(d)** eat other animals

4. In the food chain grass, deer, tiger, the primary consumer is: 1

(a) grass **(b)** bacteria **(c)** deer **(d)** tiger

5. Burning of fossil fuels increases the amount of ___ in air. 1

(a) oxygen **(b)** carbon dioxide **(c)** nitrogen **(d)** helium

6. An animal that eats only plants is a: 1

(a) omnivore **(b)** herbivore **(c)** carnivore **(d)** decomposer

7. In the food chain grass, deer, tiger, the top carnivore is: 1

(a) deer **(b)** tiger **(c)** grass **(d)** fungi

8. Water returns to the atmosphere from plants by: 1

(a) pollination **(b)** photosynthesis **(c)** absorption **(d)** transpiration

9. An animal that eats both plants and animals is a: 1

(a) producer **(b)** omnivore **(c)** herbivore **(d)** carnivore

10. Energy flows in a food chain: 1

(a) from producers to consumers **(b)** from decomposers to the Sun
(c) in a cycle **(d)** from consumers to producers

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write a food chain of a grassland having four organisms. 2

12. Name two causes of loss of biodiversity. 2

13. What are producers? Why are they important? 2

14. What is meant by a trophic level? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the flow of energy in an ecosystem. Why is it called one-way? 3

16. State three ways in which humans can protect the environment. 3

17. Write three examples of interdependence among organisms. 3

OR

Explain the 10 per cent rule with an example. 3

18. Explain the difference between a herbivore, a carnivore and an omnivore with examples. 3

OR

Explain how pollution can disturb an aquatic ecosystem. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the flow of energy through the trophic levels using a pyramid and the 10 per cent rule. Calculate energy at each level starting with 10000 J in producers. 5

OR

Explain the importance of biodiversity and measures for its conservation. 5

20. Explain the carbon cycle with the processes involved. 5

OR

Explain the water cycle and its importance for living organisms with five processes. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND